

Exercise

1. Assume that the time taken to bake a batch of cookies is normally distributed with mean 45 minutes and standard deviation 2 minutes.
 - i. Generate a random sample of size 25 for the baking time.
 - ii. Test whether the average baking time is less than 46 minutes at a 5% level of significance.

```
> #random sample of size 25 for the baking time  
> sample <- rnorm(25, mean =45, sd =2)  
> # Test whether the average baking time is less than 46 minutes at a 5%  
level of significance.  
> t.test (sample, mu=46, alternative = "less")
```

One Sample t-test

```
data: sample  
t = -2.4419, df = 24, p-value = 0.01118  
alternative hypothesis: true mean is less than 46  
95 percent confidence interval:  
-Inf 45.69418  
sample estimates:  
mean of x  
44.97842
```